

AGRI-BRIDGE

Team Members:

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Project Title: - Agri-Bridge: An AI-Enabled Multilingual Circular Economy Platform for Small-Scale Farmers and Industries

Problem Statement: - Small-scale farmers in rural India lack access to timely crop recommendations, weather-based farming advice, and real-time market prices. Relying heavily on middlemen for price information leads to significant financial loss. Without digital tools tailored to their needs, farmers make uninformed decisions on what to grow, when to sell, and how to protect crops—resulting in lower yields, increased food waste, and financial instability. Compounding this economic struggle, farmers lack direct channels to monetize crop residue, leaving them with no choice but to engage in open-field stubble burning. Every year, these burning releases massive amounts of carbon dioxide (CO₂), particulate matter (PM_{2.5}), and greenhouse gases into the atmosphere, causing severe seasonal air pollution, degrading soil health, and intensifying the regional climate crisis. There is an urgent need for an accessible, multilingual digital ecosystem that empowers farmers with actionable intelligence and connects them directly to a circular economy to upcycle agri-waste.

Abstract: -Open-field agricultural residue burning (commonly known as stubble burning) is an urgent environmental crisis in many developing nations, contributing to severe winter smog, respiratory health emergencies, and massive greenhouse gas emissions. At the same time, rural farmers struggle with financial instability, volatile market prices, and a lack of access to crop-disease advisories.

Agri-Bridge is a mobile-first web application designed to solve these issues simultaneously by establishing a decentralized, digital marketplace that links farmers directly with industries. Farmers list their excess crop residues, and the system automatically calculates the potential profits and the amount of carbon dioxide (CO₂) prevented from entering the atmosphere. Industrial buyers can browse regional listings, calculate transportation distances, and make direct offers to secure materials for bioenergy, composting, or packaging.

To ensure adoption among rural demographics, the platform breaks down technical barriers through multilingual localization (supporting English, Hindi, Tamil, Telugu, Kannada, and Malayalam) and a floating Global Voice Navigation system** powered by the Web Speech API, allowing hands-free usage. Furthermore, the application includes a simulated AI Plant Disease Scanner with high-fidelity canvas animations to guide farmers through leaf diagnosis, and a gamified Carbon Credit Leaderboard to reward eco-friendly farmers. Agri-Bridges the gap between agricultural viability and environmental sustainability

Tools and Technologies Used

- **Frontend Framework:** React.js with Vite for fast rendering and efficient development.
- **Styling & UI:** CSS3 with responsive layouts, custom themes, glass-morphism effects, and interactive animations.
- **Web APIs:** Web Speech API for voice navigation and typing; File-Reader and Media Device APIs for image uploads and camera access.
- **Database & Logic:** JSON-based datasets and rule-based logic for crop recommendations, market data, translations, and carbon calculations.
- **Hosting & Deployment:** Netlify for automated deployment, secure HTTPS hosting, and continuous integration.

Current State of the Project

The application is fully operational as an interactive multi-role web platform supporting **Farmers** and **Industrial Buyers**, localizing content across **6 languages** (English, Hindi, Tamil, Telugu, Kannada, and Malayalam).

Completed Work:

1. **P2P Marketplace:** Farmers can list waste and view estimated profits/carbon offsets. Industries can browse listings by proximity, view details, and issue deal offers.
2. **Global Voice Navigation:** A floating microphone button allows farmers to speak commands (e.g., `"Market"`, `"Prices"`, `"Scanner"`, `"Leaderboard"`) to instantly switch tabs.
3. **AI Plant Doctor:** A leaf scanner that simulates computer vision. Farmers upload/snap leaf photos, triggering a green laser scanning effect that diagnoses the infection and outputs organic remedies.
4. **Carbon Credit Leaderboard:** Gamifies stubble burning prevention. Ranks top-performing regional farmers by CO₂ tonnes offset, awarding badges (e.g., `"Eco Champion"`).
5. **AI Chatbot with Voice Typing:** A multilingual chatbot with an integrated mic that translates spoken questions into text input.
6. **Agri-Portals:** Dynamic crop recommendations based on farm inputs, live mandi price listings with rise/fall indicators, weather alerts, and direct government schemes application links.